

Session 1: Introduction

Heiner Salomon

Bremen University

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Overview I

- 1 Getting started
- 2 Getting to know each other
- 3 Introduction to course
- 4 Any questions?
- 5 Installing R
- 6 Installing Quarto
- 7 git
- 8 Any questions?

Getting started

Why are we here

- Quantitative analysis skills important (in academia and beyond)

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- Many courses teach statistics, few teach how to do the programming

Why are we here

- Quantitative analysis skills important (in academia and beyond)
- Many courses teach statistics, few teach how to do the programming
- Manual copying of data/ results always (!) error prone



Introduction to course

- I am only a facilitator

Introduction to course

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- You decide what you take from it

Introduction to course

- I am only a facilitator
- You decide what you take from it
- Idea is to work continuously on project

Learning goals of the course

- Learning some R



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Learning goals of the course

- Learning some R
- Apply what you know in real research project



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Learning goals of the course

- Learning some R
- Apply what you know in real research project
- Empower you to continue on your journey

Learning goals of today

- Introduction (you, me, course)

Learning goals of today

- Introduction (you, me, course)
- Setting expectations

Learning goals of today

- Introduction (you, me, course)
- Setting expectations
- Setting up R, Quarto and Github



Getting to know each other

Introduction template

■ Name

Introduction template

- Name
- Background

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- Name
- Background
- Knowledge of R with examples

Introduction template

- Name
- Background
- Knowledge of R with examples
- Article/ book you read recently

Introduction template

- Name
- Background
- Knowledge of R with examples
- Article/ book you read recently
- What do you want to get out of this course?



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My introduction

- Heiner Salomon

My introduction

- Heiner Salomon
- PhD researcher in Bremen



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- I have worked with R the past 3 years extensively

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- I hope to teach something and have a good time



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Introduction to course

Meetings

- We meet every Monday 14:00-16:00

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- First half of class: Presentation by me

Meetings

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- First half of class: Presentation by me
- Second half: Class discussion

Course schedule

- ~~April 6th:~~ Easter

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- June 29th: Presentations

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- June 22nd: More data visualisation in ggplot
- June 29th: Presentations
- July 6th: Final class

Assessment

- Progress report (Every two weeks Friday at 15:00)

3 Credit Points (CP)

6 Credit Points (CP)

tät

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- Simple progress report, free form (one short paragraph)
- I tried setting it up on StudIP, might change
- I expect 5 short reports:
 - one can be dropped without explanation
 - after that need a good excuse



Assessment - Script (each 25%)

- Correct: Is the script correct and does what it should?

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- Comments: Are the comments in the script enough to follow it?

Assessment - Presentation criteria (each 25%)

- Structure: Is the structure helpful and appropriate?

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- Clarity: Does it become clear what you present on?
- Visuals: Does the presentation look appealing?
- Presentation: Are you presenting convincingly?



Assessment - Research note criteria (each 25%)

- Structure: Is the structure helpful and appropriate?

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Assessment - Research note criteria (each 25%)

- Structure: Is the structure helpful and appropriate?
- Clarity: Does it become clear what you are writing on?
- Visuals: Does the data visualisation look appealing?
- References: Are the references correct and relevant?



Etiquette

- Attendance is not mandatory, but highly encouraged

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- You can always contact me at heiner.salomon@uni-bremen.de

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- You can always contact me at heiner.salomon@uni-bremen.de
- Communicate early, communicate often

Where to find help

- This introduction is great: <https://r4ds.hadley.nz/intro.html>

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- You can also try Large Language Models



Use of LLMs

- Large Language Models can be extremely helpful for coding

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- But they also have limits

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- You chose this course because you want to learn

Use of LLMs

- Large Language Models can be extremely helpful for coding
- But they also have limits
- You chose this course because you want to learn
- “People go to the gym even though we have forklifts”

Any questions?

Installing R



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Why R?

Pro

- free

Contra



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Why R?

Pro

- free
- open source

Contra



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Why R?

Pro

- free
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Contra



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- steeper learning curve



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Why R?

Pro

- free
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Contra

- steeper learning curve
- many ways to achieve the same goal



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Why R?

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- free
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- steeper learning curve
- many ways to achieve the same goal
- No authoratative singular guide online



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- If you use R, one should use RStudio
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- Adds features that make programming easier
- Unfortunately, proprietary (but for free)



Installing R and RStudio

- Download R here: <https://cran.r-project.org/mirrors.html>

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- Download RStudio here:
<https://posit.co/download/rstudio-desktop>

Installing Quarto



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- Same (for R-users) as RMarkdown
- Seamless integrates (R-)code and text with references (Zotero)
- Outputs PDFs, but also HTML (and other formats)



Do you need to use Quarto?

- No, but it is highly encouraged

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- We will be working in R
- Adding Quarto costs little extra effort but adds a lot
- Very good explanatory resources here:
<https://quarto.org/docs/guide/>

Installing Quarto

- You can download Quarto here:
<https://quarto.org/docs/get-started/>

git

Why git?

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- git is a distributed version control software system
- Allows to save and track progress in the cloud
- Enables collaboration on code with a team
- There are a range of providers for git (e.g. www.github.com)

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- It costs a little bit of time to set up

Do you need to use git?

- No, but it is encouraged
- It costs a little bit of time to set up
- but once set-up, it acts as a safety net for your code

Installing and setting up git

- Great explainer on how to integrate git in R:
<https://happygitwithr.com/>

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- another explainer: <https://rfortherestofus.com/2021/02/how-to-use-git-github-with-r/>

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